Auto Ticket Classification Using Flow Designer

This is a good project for your BCA and ServiceNow practical work. You can build it using **ServiceNow Flow Designer**, which automates ticket handling without requiring you to write a large amount of code.

1. Project overview

Project title: Auto Ticket Classification Using Flow Designer

Domain: IT Service Management (ITSM) and Workflow Automation

Platform: ServiceNow

Main objective: To automatically classify support tickets based on their description or keywords, assign the correct category and priority, and route them to the appropriate support team using Flow Designer.

Example

Suppose an employee creates a ticket:

"My laptop is not turning on."

The system automatically identifies it as a hardware issue, assigns the Hardware category, sets a suitable priority, and routes the ticket to the IT support team.

Other examples:

Ticket description	Classification	Assignment team
Email is not working	Software / Email	Application Support
Laptop screen is broken	Hardware	Hardware Support
Cannot connect to Wi-Fi	Network	Network Support
Password reset required	Access / Account	IT Help Desk

The exact categories and assignment groups depend on how you configure your ServiceNow instance.

2. Project objectives

- Automatically classify tickets based on keywords.
- Set the ticket category and subcategory.
- Assign priority based on predefined conditions.
- Route tickets to the correct support group.
- Reduce manual ticket assignment.

- Improve response time and ticket management.
- Maintain a record of ticket status and assignment.

3. Modules of the project

Module 1: Ticket Creation

Employees create incident tickets by entering a short description, detailed description, and other required information.

Module 2: Automatic Classification

The flow checks ticket keywords and identifies categories such as Hardware, Software, Network, or Account Access.

Module 3: Automatic Assignment

The system selects the appropriate assignment group according to the classification rules.

Module 4: Priority Management

The flow assigns priority using predefined rules, such as treating critical service outages differently from routine requests.

Module 5: Tracking and Reporting

Administrators monitor ticket status, category, priority, assignment group, and the number of tickets classified automatically.

4. Software and tools required

Tool	Purpose
ServiceNow instance	Platform for managing incident tickets
Flow Designer	Creates automation workflows
Incident table	Stores ticket details
Conditions and branches	Classify tickets based on keywords
Update Record action	Updates category, priority, and assignment
Reports	Displays ticket statistics

You can implement the main project using ServiceNow's configuration tools rather than HTML, CSS, Python, or Django.

5. Workflow design

This is the main workflow you will create in Flow Designer.

Trigger: Incident Created

A new ticket is submitted

Check Short Description

Evaluate ticket keywords

Hardware

Hardware Support

Network

Network Support

Software

Application Support

Account Access

IT Help Desk

Update Incident Record

Set category, priority, and assignment group

Ticket Classified

Ticket is ready for the support team

Note: These are the proposed project rules, not a live ServiceNow flow. You will configure and test each branch in your instance.

6. How to implement it in ServiceNow

Follow these steps in order.

1. **Open Flow Designer**

Log in to your ServiceNow instance. Search for **Flow Designer** in the Application Navigator and open it. Depending on your instance version, it may be available under Workflow Studio.

2. **Create a new flow**

   Select **New → Flow**. Enter the name `Auto Ticket Classification`, add a description, and select the appropriate application scope.

3. **Configure the trigger**

   Choose the trigger type **Record**. Select the **Incident incidentincident** table and configure the trigger to run when a record is **Created**.

4. **Add classification conditions**

   Add an **If** condition to check the Short description field. For example, if it contains `laptop`, classify it as Hardware. Add Else If branches for network, email/software, and password-related problems.

5. **Add Update Record actions**

   For each classification branch, use the Update Record action to update the triggering incident's category and assignment group. Set the appropriate priority where your rules require it.

6. **Handle unmatched tickets**

   Add a default branch for descriptions that do not match your keywords. Keep the existing category where appropriate or route the ticket to a general IT Help Desk group for manual review.

7. **Save, activate, and test**

   Save and activate the flow. Create test incidents with different descriptions and verify that the category, priority, and assignment group are updated correctly.

Important: Use the actual assignment groups available in your instance. The flow must also avoid repeatedly triggering itself through its own record updates; configuring the trigger for creation only helps prevent that issue.

7. Test cases

Test input	Expected category	Expected result
Laptop is not turning on	Hardware	Assigned to Hardware Support
Wi-Fi connection failed	Network	Assigned to Network Support
Email application is not working	Software	Assigned to Application

Test input	Expected category	Expected result
		Support
Please reset my password	Account Access	Assigned to IT Help Desk
General request with no matching keyword	Default	Routed for manual review

These are expected results. You should record the actual results after testing your flow.

8. Project abstract

Abstract

The Auto Ticket Classification Using Flow Designer project aims to automate the classification and assignment of IT support tickets using the ServiceNow platform. In traditional ticket management systems, support teams often classify incidents and assign them to the appropriate departments manually. This process can consume time, create assignment errors, and delay incident resolution.

The proposed system uses ServiceNow Flow Designer to automate ticket processing. When a new incident is created, the flow evaluates its description against predefined keywords and conditions. Based on the matching rules, the system assigns a suitable category, priority, and support group.

The project includes ticket creation, automatic classification, assignment management, priority handling, and ticket tracking. It also provides a default handling process for tickets that do not match predefined rules.

The system aims to reduce repetitive manual work, improve ticket routing accuracy, and increase the efficiency of IT service management. It can be extended in the future with natural language processing and machine learning to classify tickets using their meaning rather than predefined keywords.

9. Expected outcome

By completing this project, you should be able to demonstrate:

- A working ServiceNow Flow Designer flow.
- Automatic incident categorization using rules.
- Automatic assignment to configured support groups.
- Priority handling based on conditions.
- Test evidence showing the before-and-after incident records.
- Reports summarizing ticket classifications.